A Resource Theory of the Non-Hermitian Skin Effect

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Abstract

The non-Hermitian skin effect (NHSE), the accumulation of an extensive number of eigenmodes at a boundary, is usually diagnosed model by model. We formulate it as a quantum resource theory whose objects are the right and left eigen-responses of an arbitrary finite-dimensional generator, with boundary collars, in any spatial dimension. The resource has two layers. *Accumulation*, the NHSE itself, is the violation of Bessel's inequality by the right responses on a boundary collar, quantified by a collar Bessel bound B. *Direction*, right responses at one boundary and left responses at the opposite one, is quantified by the advantage s of a boundary-only test in discriminating the two. We prove the following. (i) Reciprocity ($G^T = UGU^\dagger$ with a boundary-compatible unitary U) forbids direction for every diagonalizable generator but not accumulation, so the $\mathbb{Z}_2$ and geometry-dependent skin effects are accumulation without direction. (ii) Extensive direction forces accumulation, accumulation forces an ill-conditioned eigenbasis, and direction forces $\log \kappa_m \geq s_m - \frac{1}{2}\log 2$ for the eigenvalue condition numbers and a large directional resolvent residue. (iii) The free operations apply one collar-non-expanding channel to every response, process mode labels classically, and never raise the density of modes in a collar. They form a symmetric monoidal category, create neither layer, admit a complete family of monotones computable by semidefinite programming, and on generators are exactly embeddings of a function of the generator as a decoupled block. (iv) Directional rates are totally ordered, and Hatano–Nelson chains are mutually inconvertible at fixed size. (v) For tridiagonal generators with positive hopping products, clean or disordered, both layers are bounded below by the accumulated imaginary gauge field. Beyond tridiagonal chains, lower bounds rest on explicit hypotheses imported from non-Bloch band theory.

1 Introduction

Under open boundary conditions, a non-Hermitian generator can localize an extensive fraction of its eigenvectors at a boundary: the non-Hermitian skin effect (NHSE) [1, 2, 3, 4, 5, 6, 7, 8], reviewed in [9, 10]. Skin modes accompany a non-zero point-gap winding of the periodic-boundary spectrum in one dimension [6, 7] and a non-zero spectral area in higher dimensions [8]. Non-reciprocity is not required: the $\mathbb{Z}_2$ skin effect [6] and the geometry-dependent skin effect of reciprocal lattices [8, 30] have no preferred direction. The consequences include exponential spectral sensitivity [12, 14], large Petermann factors [15], and directional amplification [16, 17].

Quantum resource theories [21, 22] organize a property around free objects, free operations that cannot create it, and monotones that decide conversions. Our design goals are: (U1) every finite-dimensional generator with a boundary structure defines an object, in any dimension and for any lattice, disorder or internal structure; (U2) every normal generator is free, and every form of the NHSE, reciprocal or not, is a resource; (U3) the free operations never create the resource, admit a *complete* family of monotones, and are physical operations on generators; (U4) the resource controls measurable quantities and is calibrated on standard models with explicit constants. Sections 2–7 establish (U1)–(U4), and Section 9 states what is not established.

The objects are pairs of classical–quantum states, ω^R and ω^L, processed by a common channel, as in the resource theory of asymmetric distinguishability [25] and the comparison of statistical experiments [26, 27]. Accumulation compares the right responses with an orthonormal family through Bessel's inequality; direction is the part of the right/left distinguishability that a boundary-only test can read; reciprocity removes direction but not accumulation (Theorem 2). Related resource theories treat the antiunitary asymmetry of states [38] and the time-reversal violation of channels [39]; here the objects are eigen-data of non-Hermitian generators.

2 Objects

Throughout, Σ is a finite set of boundary labels with a fixed-point-free involution ι that assigns to each boundary its opposite. In one dimension $\Sigma = \{+, -\}$ (right and left ends) and $\iota(\pm) = \mp$. In d dimensions $\Sigma = \{\pm x_1, \dots, \pm x_d\}$, and corners can be included. A width cutoff $W \in \mathbb{N}$ is fixed at each system size.

Definition 1 (Skin datum)**.** A skin datum X consists of:

- a finite-dimensional Hilbert space H;
- projectors $\Pi_{\sigma,w}$ on H for $\sigma \in \Sigma$ and $w \in [W]$, nested in w (the *collars*);
- M responses, each of weight $\pi_m = 1/M$;
- density operators ρ^R_m, ρ^L_m on H (the right and left responses).

X is *boundary-separated* if $\Pi_{\sigma,W}\Pi_{\iota\sigma,W} = 0$ for all σ, and then, by nesting, $\Pi_{\sigma,w}\Pi_{\iota\sigma,w} = 0$ for all w. The class of boundary-separated skin data is denoted $\mathfrak{O}$. The *type* of X is $(M, H, \{\Pi_{\sigma,w}\})$. The classical–quantum (cq) operators are $\omega^h_X = \sum_m \pi_m |m\rangle\langle m| \otimes \rho^h_m$ for $h \in \{R, L\}$, and $\mathbb{E}_\pi$ denotes the average over the responses.

Equal weights make every eigenmode count once; this is what the number of modes in Definition 4 refers to.

Definition 2 (Generator datum)**.** Let G be a matrix on $H = \ell^2(\mathcal{C}) \otimes \mathbb{C}^d$, where $\mathcal{C}$ is a finite set of cells with open boundaries, and let $\Pi_{\sigma,w}$ project onto the w cell layers nearest boundary σ. For a semisimple eigenvalue λ of multiplicity k with spectral projector P_λ, let Q^R_λ and Q^L_λ be the orthogonal projectors onto $\operatorname{ran} P_\lambda$ and $\operatorname{ran} P^\dagger_\lambda$, the right and left eigenspaces. The generator datum $X_\Lambda(G)$ of a set Λ of semisimple eigenvalues has, for each $\lambda \in \Lambda$, k responses with $\rho^R = Q^R_\lambda/k$ and $\rho^L = Q^L_\lambda/k$. For a simple eigenvalue, $\rho^R = |R\rangle\langle R|$ and $\rho^L = |L\rangle\langle L|$ with unit vectors $GR = \lambda R$ and $G^\dagger L = \bar{\lambda} L$, so that $L^\dagger R \neq 0$. We write $X(G)$ when G is diagonalizable and Λ is its whole spectrum. Liouvillian, Floquet and scattering generators enter the same way.

The data do not depend on bases inside eigenspaces, and exceptional points (non-semisimple eigenvalues) are excluded. A POVM read-out $\{E_x\}$ of the boundary becomes projective under the isometry $\psi \mapsto \sum_x \sqrt{E_x}\psi \otimes |x\rangle$, which leaves every fidelity and divergence below invariant, so collars may be taken projective.

3 Two layers: direction and accumulation

Definition 3 (Paired leakage and directional score)**.** For a response m, boundary σ and width w, the paired leakages are

$$\lambda_m^R(\sigma, w) = \mathrm{Tr}[(\mathbb{I} - \Pi_{\sigma,w})\rho_m^R], \qquad \lambda_m^L(\sigma, w) = \mathrm{Tr}[(\mathbb{I} - \Pi_{\iota\sigma,w})\rho_m^L].$$

The right response is tested at σ and the left response at the opposite boundary $\iota\sigma$. The directional score is

$$s_m(\sigma, w) = \tfrac{1}{2}\log^+ \frac{1}{\lambda_m^R(\sigma, w) + \lambda_m^L(\sigma, w)} \in [0, \infty], \qquad \log^+ x = \max(0, \log x),$$

and its average is $\mathcal{R}_\sigma(w) = \mathbb{E}_\pi\, s_m(\sigma, w)$.

Remark 1 (Operational meaning). Given a copy of ρ_m^R or ρ_m^L with equal priors, perform the boundary-only test $\{\Pi_{\sigma,w}, \Pi_{\iota\sigma,w}, \mathbb{I} - \Pi_{\sigma,w} - \Pi_{\iota\sigma,w}\}$. Answer "$R$" on the first outcome, "L" on the second, and abstain otherwise. The probability of a correct conclusive answer is $P_{\mathrm{succ}} = 1 - \frac{1}{2}(\lambda_m^R + \lambda_m^L)$, so

$$s_m = \tfrac{1}{2}\log^+ \frac{1}{2(1 - P_{\mathrm{succ}})}.$$

If the two responses coincide, $P_{\mathrm{succ}} \le \frac{1}{2}$. Thus $s_m > 0$ exactly when the boundary alone discriminates the right response from the left one better than it could discriminate two copies of the same state.

Definition 4 (Accumulation)**.** The *right frame* of X is $S_X = \sum_m \rho_m^R$, and the *collar Bessel bound* is

$$B_\sigma(w) = \left\|\Pi_{\sigma,w} S_X \Pi_{\sigma,w}\right\|.$$

For a unit vector v in a collar, $\langle v|S_X|v\rangle = \sum_m \langle v|\rho_m^R|v\rangle$. Orthonormal pure responses satisfy $S_X \le \mathbb{I}$ (Bessel's inequality), so $B_\sigma(w) > 1$ certifies that the right responses crowd the collar more than any orthonormal family can: a detector for v shown the right response of a uniformly random mode clicks with probability up to $B_\sigma(w)/M$, against $1/M$ for orthonormal responses. The NHSE is $B_\sigma(w)$ growing with system size; non-normal generators without a skin effect give $B = O(1)$ (Remark 3).

Definition 5 (Free objects)**.** $\mathcal{F}_{\mathrm{dir}} = \{X \in \mathfrak{O} :\ s_m(\sigma, w) = 0\ \forall m, \sigma, w\}$, $\mathcal{F}_{\mathrm{acc}} = \{X \in \mathfrak{O} :\ B_\sigma(w) \le 1\ \forall \sigma, w\}$, and $\mathcal{F} = \mathcal{F}_{\mathrm{dir}} \cap \mathcal{F}_{\mathrm{acc}}$. Equivalently, $X \in \mathcal{F}_{\mathrm{dir}}$ if and only if $\lambda_m^R + \lambda_m^L \ge 1$ for all m, σ, w.

Lemma 1 (Collar reciprocity implies no direction)**.** *If $X \in \mathfrak{O}$ and $\mathrm{Tr}[\Pi_{\sigma,w}\rho_m^R] = \mathrm{Tr}[\Pi_{\sigma,w}\rho_m^L]$ for all m, σ, w, then $X \in \mathcal{F}_{\mathrm{dir}}$.*

Proof. $\lambda_m^R + \lambda_m^L = 2 - \mathrm{Tr}[\Pi_{\sigma,w}\rho_m^L] - \mathrm{Tr}[\Pi_{\iota\sigma,w}\rho_m^L] \ge 1$, because $\Pi_{\sigma,w} + \Pi_{\iota\sigma,w}$ is a projector. □

Theorem 2 (Reciprocity)**.** *Let G be as in Definition 2, with transposition taken in a basis in which every $\Pi_{\sigma,w}$ is real. Suppose $G^T = UGU^\dagger$ for a unitary U that commutes with every $\Pi_{\sigma,w}$ (for example, a cell-local unitary).*

(a) For every semisimple eigenvalue λ, $Q_\lambda^L = \overline{UQ_\lambda^R U^\dagger}$, hence $\mathrm{Tr}[\Pi_{\sigma,w}Q_\lambda^R] = \mathrm{Tr}[\Pi_{\sigma,w}Q_\lambda^L]$ for all σ, w. Every generator datum of G lies in $\mathcal{F}_{\mathrm{dir}}$.

(b) If $U^T = -U$, every semisimple eigenvalue of G has even multiplicity.

Proof. For a semisimple λ of multiplicity k, the λ-eigenspaces of G and G^T and the $\bar{\lambda}$-eigenspace of $G^\dagger$ all have dimension k, and $G^T v = \lambda v$ holds if and only if $G^\dagger \bar{v} = \bar{\lambda}\bar{v}$. For $E = \operatorname{ran} P_\lambda$, $G^T(UE) = UGE = \lambda UE$, so UE is the λ-eigenspace of G^T and $\operatorname{ran} P_\lambda^\dagger = \overline{UE}$. (a) Hence $Q_\lambda^L = \overline{UQ_\lambda^R U^\dagger}$, and $\operatorname{Tr}[\Pi Q_\lambda^L] = \overline{\operatorname{Tr}[\Pi U Q_\lambda^R U^\dagger]} = \operatorname{Tr}[\Pi Q_\lambda^R]$ because Π is real and commutes with U. Lemma 1 gives the rest. (b) The pairing $(l, r) \mapsto l^\dagger r$ between $\operatorname{ran} P_\lambda^\dagger$ and E is non-degenerate, since $l = P_\lambda^\dagger x$ gives $l^\dagger r = x^\dagger r$ for $r \in E$. With $l = \overline{Ur'}$ and $r' \in E$ it becomes the bilinear form $r'^T U^T r$ on E, which is antisymmetric if $U^T = -U$; a non-degenerate antisymmetric form exists only in even dimension. □

Corollary 3. *Data of normal generators, in particular Hermitian ones, lie in $\mathcal{F}$: their eigenspaces are mutually orthogonal and $Q_\lambda^L = Q_\lambda^R$, so $S_X \le \mathbb{I}$ and $\rho_m^R = \rho_m^L$. Data of generators that are reciprocal up to a cell-local gauge, such as complex-symmetric gain/loss lattices, lie in $\mathcal{F}_{\mathrm{dir}}$ but need not lie in $\mathcal{F}_{\mathrm{acc}}$.*

Remark 2 (Reciprocal skin effects). Reciprocity with $U^T = -U$ is the transpose-type time reversal with $TT^* = -1$ [11] that protects the $\mathbb{Z}_2$ skin effect [6]. Theorem 2 shows that its degeneracy is forced and that it carries no direction, and Corollary 19 shows that it accumulates. The geometry-dependent skin effect of reciprocal lattices [8, 30] (Section 7) and the bidirectional skin effect [31] also fall under Theorem 2.

Theorem 4 (Direction forces spectral sensitivity)**.** *Let $X \in \mathfrak{O}$ and fix m, σ, w, abbreviating $\lambda^h = \lambda_m^h(\sigma, w)$ and $s = s_m(\sigma, w)$.*

(a) $F(\rho_m^R, \rho_m^L) \le \sqrt{\lambda^R} + \sqrt{\lambda^L} \le \sqrt{2}\, e^{-s}$, where $F(\rho, \tau) = \|\sqrt{\rho}\sqrt{\tau}\|_1$. Hence

$$\Phi_Q(X) := -\log \sum_m \pi_m F(\rho_m^R, \rho_m^L) \;\ge\; -\log \mathbb{E}_\pi e^{-s_m(\sigma,w)} - \tfrac{1}{2}\log 2.$$

(b) For a simple eigenvalue λ_m of a generator, let $P_m = R_m L_m^\dagger / (L_m^\dagger R_m)$ be the spectral projector of λ_m, and let $\kappa_m = \|P_m\| = |L_m^\dagger R_m|^{-1}$ be the eigenvalue condition number [13, 12]. Then

$$\log \kappa_m \;\ge\; s - \tfrac{1}{2}\log 2, \qquad \|\Pi_{\sigma,w} P_m \Pi_{\iota\sigma,w}\| = \kappa_m \sqrt{(1-\lambda^R)(1-\lambda^L)} \;\ge\; \sqrt{\tfrac{1}{2}(e^{2s}-1)},$$

and, if $s > 0$, $\|\Pi_{\iota\sigma,w} P_m \Pi_{\sigma,w}\| \le \frac{1}{2}\kappa_m e^{-2s}$.

Proof. (a) Because the collars are orthogonal, Remark 1 defines a valid three-outcome measurement. Its outcome laws are $p = (p_1, p_2, p_3)$ and $q = (q_1, q_2, q_3)$. Fidelity does not decrease under measurement [28], so $F \le \sum_i \sqrt{p_i q_i}$. Here $p_2 + p_3 = \lambda^R$ and $q_1 + q_3 = \lambda^L$. Cauchy–Schwarz gives $\sqrt{p_1 q_1} + \sqrt{p_3 q_3} \le \sqrt{(p_1 + p_3)(q_1 + q_3)} \le \sqrt{\lambda^L}$, and $\sqrt{p_2 q_2} \le \sqrt{\lambda^R}$. Next, $\sqrt{\lambda^R} + \sqrt{\lambda^L} \le \sqrt{2(\lambda^R + \lambda^L)}$, which equals $\sqrt{2}e^{-s}$ when $s > 0$; when $s = 0$ the bound $F \le 1 \le \sqrt{2}$ is trivial. Averaging over m gives the bound on Φ_Q. (b) For pure responses $F = |L_m^\dagger R_m| = \kappa_m^{-1}$, so the first inequality is (a). $\Pi_{\sigma,w} P_m \Pi_{\iota\sigma,w}$ has rank one with norm $\|\Pi_{\sigma,w} R_m\|\, \|\Pi_{\iota\sigma,w} L_m\|\, \kappa_m$, which gives the identity. Using $(1-\lambda^R)(1-\lambda^L) \ge 1 - e^{-2s}$ and $\kappa_m \ge e^s/\sqrt{2}$ gives the lower bound. For the reverse block, $\|\Pi_{\iota\sigma,w} R_m\|^2 \le \lambda^R$ and $\|\Pi_{\sigma,w} L_m\|^2 \le \lambda^L$, so it is at most $\kappa_m \sqrt{\lambda^R \lambda^L} \le \kappa_m(\lambda^R + \lambda^L)/2$. □

Theorem 4(b) is the physical content of the directional score. Near λ_m, $(z - G)^{-1} = P_m/(z - \lambda_m) + O(1)$, so $\Pi_{\sigma,w} P_m \Pi_{\iota\sigma,w}$ is the response at boundary σ to a drive at the opposite boundary. It exceeds $\sqrt{(e^{2s}-1)/2}$ and beats the reverse block by a factor of at least $2e^{2s}\sqrt{1 - e^{-2s}}$, the mechanism of directional amplification [16, 17]. Since $|\delta\lambda_m| \le \kappa_m \|\delta G\| + O(\|\delta G\|^2)$ and κ_m^2 is the Petermann factor [15], a score s guarantees a sensitivity of order e^s [14]. For the Hatano–Nelson (HN) chain, $\log \kappa_j \simeq gL$ and $\max_{w \le L/2} s_j \simeq gL/2$.

Theorem 5 (Direction, accumulation and non-normality)**.** *Let $X \in \mathfrak{O}$, fix σ, w, and let $r = \operatorname{rank}\Pi_{\sigma,w}$.*

(a) $B_\sigma(w) \geq \frac{1}{r}\sum_m \big(1 - e^{-2s_m(\sigma,w)}\big)$. Extensive direction therefore forces accumulation: if a fraction f of the responses has $s_m(\sigma,w) \geq \eta > 0$, then $B_\sigma(w) \geq fM(1-e^{-2\eta})/r$.

(b) Let $X = X(G)$ for a diagonalizable G with simple spectrum on H, $\dim H = M$, and let $E = \operatorname{Tr}[\Pi_{\sigma,w} S_X] - r$ be the collar excess. If $r < M$, then

$$\textstyle\sum_m \kappa_m^2 \;\geq\; \frac{M-r}{M-r-E}.$$

Proof. (a) $B_\sigma(w) \geq \operatorname{Tr}[\Pi S_X \Pi]/r = \frac{1}{r}\sum_m (1-\lambda_m^R)$, and $\lambda_m^R \leq \lambda_m^R + \lambda_m^L = e^{-2s_m}$ when $s_m > 0$. (b) Let R be the matrix with columns R_m. Then $S_X = RR^\dagger$ and $\operatorname{Tr} S_X = M$. The smallest eigenvalue of S_X is at most its average over $\operatorname{ran}(\mathbb{I}-\Pi)$, which is $\operatorname{Tr}[(\mathbb{I}-\Pi)S_X]/(M-r) = (M-r-E)/(M-r)$. The rows of R^{-1} are $L_m^\dagger/(L_m^\dagger R_m)$, of norms κ_m, so $\sum_m \kappa_m^2 = \|R^{-1}\|_F^2 \geq \|R^{-1}\|^2 = 1/\lambda_{\min}(S_X)$. □

Accumulation is thus a collective form of non-normality: it forces an ill-conditioned eigenbasis, but, unlike direction (Theorem 4), not a large condition number for every mode.

Remark 3 (The converse fails). Non-normality does not imply skin. A Hermitian chain with one strongly non-reciprocal bulk bond (hoppings 3 and 0.2) has mean $\log\kappa_m \approx 0.68$ at every size but $\kappa_m \leq 2.07$, so Theorem 4(b) caps the score at $s_m < 1.1$ for all $w \leq L/2$ (V32-06, $L \leq 160$), and its collar Bessel bound stays below 1.9 (V32-12): neither layer grows. Theorem 4 is a one-way bridge from direction to non-normality.

Definition 6 (Skin accumulation)**.** A sequence (X_L) with widths w_L ($2w_L$ at most the linear size) shows *directed skin accumulation* at boundary σ and scale a (with $a(w) \to \infty$) if $\mathcal{R}_\sigma(w_L) \geq \eta\, a(w_L)$ for some $\eta > 0$ and all large L. The accumulation is *extensive* if moreover $\pi\{m : s_m(\sigma, w_L) \geq \eta\, a(w_L)\} \geq f > 0$. The rates are $r_\sigma^\pm = \limsup / \liminf_L \mathcal{R}_\sigma(w_L)/a(w_L)$. The sequence is *uniformly scaling with rate r* if $\max_m |s_m(\sigma, w_L)/a(w_L) - r| \to 0$. It shows *extensive boundary accumulation* at σ if $c_\sigma^R(w_L) = \mathbb{E}_\pi \operatorname{Tr}[\Pi_{\sigma,w_L}\rho_m^R] \geq f > 0$ while $\operatorname{rank}\Pi_{\sigma,w_L}/M_L \to 0$; then $B_\sigma(w_L) \geq M_L c_\sigma^R(w_L)/\operatorname{rank}\Pi_{\sigma,w_L} \to \infty$.

4 Free operations and monotones

Definition 7 (Free processing and free morphism)**.** Let X have type (M, H, Π) and let $\tau = (M', H', \Pi')$ be a type. A *label-blind processing* of X into τ is the map

$$\mathcal{K} \otimes \Xi : \; |m\rangle\langle m| \otimes \rho \;\mapsto\; \textstyle\sum_{m'} \nu_{m'm} |m'\rangle\langle m'| \otimes \Xi(\rho)$$

on cq operators (the label is dephased first), where $\nu \in [0,1]^{M' \times M}$ is column-stochastic and $\Xi : B(H) \to B(H')$ is a CPTP map that is *collar non-expanding*:

$$\text{(C)} \quad \Xi^\dagger(\Pi'_{\sigma,w}) \leq \Pi_{\sigma,w} \qquad \text{for all } \sigma, w.$$

A *free processing* is a finite convex combination $\Theta = \sum_j p_j\, \mathcal{K}_j \otimes \Xi_j$ of label-blind processings into τ such that $\nu = \sum_j p_j \nu_j$ maps uniform weights to uniform weights and

$$\text{(D)} \quad \tfrac{M'}{M}\, \Pi'_{\sigma,w}\, \bar{\Xi}(\mathbb{I})\, \Pi'_{\sigma,w} \leq \Pi'_{\sigma,w} \quad \text{for all } \sigma, w, \qquad \bar{\Xi} = \textstyle\sum_j p_j \Xi_j.$$

A *free morphism* $X \to Y$ is a free processing into the type of Y such that

$$\text{(M1)} \quad \Theta(\omega_X^R) = \omega_Y^R, \qquad \Theta(\omega_X^L) = \omega_Y^L.$$

We write $X \to Y$ if a free morphism exists.

By (M1) the *same* processing acts on the right and left responses, blind to whether G or $G^\dagger$ was realized [25]. By *label-blindness* one channel acts on every eigenmode, while labels are relabelled, merged or forgotten classically; an operation on the generator acts this way, and Section 5 proves the converse. By (C), *collar non-expansion*, the probability of ending in a collar never exceeds that of starting there, so no weight is pumped toward a boundary, either non-reciprocally (the origin of skin) or reciprocally (which can move an existing right/left separation inward and raise a fixed-width score). By (D), *density non-increase*, the number of modes per collar dimension never rises; this excludes coarse-graining internal levels (a partial trace satisfies (C) but doubles $\Xi(\mathbb{I})$ on a collar) and splitting a mode into several labels without enlarging the space (V32-13).

Proposition 6 (Examples of free morphisms)**.** *Each of the following is a free processing.*

(a) Boundary-compatible embeddings. $\nu = \mathbb{I}$ *and* $\Xi = V(\cdot)V^\dagger$ *for an isometry* $V : H \to H'$ *with* $V^\dagger \Pi'_{\sigma,w} V \le \Pi_{\sigma,w}$*, such as a cell-local unitary, an enlarged internal space, inserted bulk cells, or cells attached beyond a boundary (Theorem 12).*

(b) Collar-compatible channels. $\nu = \mathbb{I}$ *and any* Ξ *with (C) and* $\Pi'_{\sigma,w}\Xi(\mathbb{I})\Pi'_{\sigma,w} \le \Pi'_{\sigma,w}$*, for example cell-local unital noise or dephasing.*

(c) Response processing. $\Xi = \mathrm{id}$ *and any column-stochastic* ν *with* $M' \le M$ *that maps uniform weights to uniform weights, such as relabelling and uniform merging.*

(d) Erasure. *If* ω *is a bulk state (*$\mathrm{Tr}[\Pi_{\sigma,W}\omega] = 0$ *for all* σ*), then* $\nu = (1, \dots, 1)$ *and* $\Xi = \mathrm{Tr}(\cdot)\,\omega$ *give a free morphism* $X \to O_\omega$*, where* O_ω *has one response* $\rho^R = \rho^L = \omega$*. Moreover* $O_\omega \in \mathcal{F}$*.*

(e) Dilution. *For* $t \in [0, 1]$*,* $\nu = \mathbb{I}$ *and* $\Xi_t = (1-t)\,\mathrm{id} + t\mathrm{Tr}(\cdot)\,\omega$ *give a free morphism* $X \to X_t$*, where* X_t *has responses* $(1-t)\rho^h_m + t\omega$*.*

(f) Appending a free object. *If* $\omega^R_F = \omega^L_F$ *and* $F \in \mathcal{F}_{\mathrm{acc}}$*, then* $X \to X \otimes F$*, with the tensor product of Proposition 9.*

Proof. (C): in (a), $\Xi^\dagger(\Pi') = V^\dagger \Pi' V$, and moving a boundary outward by k cells gives $V^\dagger \Pi'_{\sigma,w} V = \Pi_{\sigma,w-k}$; (b) is the hypothesis; (c) gives equality; in (d) and (e) the adjoint of $\mathrm{Tr}(\cdot)\,\omega$, which is $A \mapsto \mathrm{Tr}[A\omega]\mathbb{I}$, vanishes on collars. (D): in (a), $\Pi' V V^\dagger \Pi' \le \Pi'$; (b) is the hypothesis; in (c), $(M'/M)\Pi' \le \Pi'$; in (d) and (e), $\Pi'\Xi_t(\mathbb{I})\Pi' = (1-t)\Pi'$. In (f), the label m goes to (m, n) with probability $1/M_F$ and $\Xi_n(\rho) = \rho \otimes \rho^R_{F,n}$; then $\Xi^\dagger_n(\Pi \otimes \Pi^F) \le \Pi$, $\bar{\Xi}(\mathbb{I}) = \mathbb{I} \otimes S_F/M_F$, (D) is the statement $F \in \mathcal{F}_{\mathrm{acc}}$, and (M1) uses $\rho^R_{F,n} = \rho^L_{F,n}$. □

Lemma 7 (Leakage domination)**.** *Let* $\Theta = \sum_j p_j \mathcal{K}_j \otimes \Xi_j : X \to Y$ *be a free morphism,* $\nu = \sum_j p_j \nu_j$*, and* $\beta_{mm'} = \nu_{m'm}\pi_m/\pi'_{m'}$*, a probability vector in* m*. Then for all* m', h, σ, w*,*

$$\lambda^{h,Y}_{m'}(\sigma, w) \;\ge\; \sum_m \beta_{mm'} \lambda^{h,X}_m(\sigma, w).$$

Proof. Let Π be $\Pi_{\sigma,w}$ for $h = R$ and $\Pi_{\iota\sigma,w}$ for $h = L$, and Π' likewise. By (M1), $\pi'_{m'}\rho'^h_{m'} = \sum_{j,m} p_j \nu^j_{m'm} \pi_m \Xi_j(\rho^h_m)$. Hence, by (C),

$$\pi'_{m'}(1 - \lambda'^h_{m'}) = \textstyle\sum_{j,m} p_j \nu^j_{m'm} \pi_m \mathrm{Tr}[\Xi^\dagger_j(\Pi')\rho^h_m] \le \sum_m \nu_{m'm}\pi_m(1 - \lambda^h_m).$$

Divide by $\pi'_{m'} = \sum_m \nu_{m'm}\pi_m$. □

Conversely, a channel that violates (C) at (σ, w) lowers $\lambda^R(\sigma, w)$ of some one-response object, so (C) is the weakest condition on a label-blind channel under which every leakage monotone (Theorem 10(ii) below) is monotone. Likewise, a processing of a type with $M = \dim H$ that violates (D) maps the complete orthonormal datum of a Hermitian generator, for which $S_X = \mathbb{I}$, to an object with $B > 1$.

Lemma 8 (Accumulation cannot grow)**.** *If $X \to Y$, then $B_\sigma(w)(Y) \le B_\sigma(w)(X)$ for all σ, w.*

Proof. With uniform weights, (M1) gives $\rho'^R_{m'} = M' \sum_{j,m} p_j \nu^j_{m'm} \pi_m \Xi_j(\rho^R_m)$, so $S_Y = \frac{M'}{M} \bar{\Xi}(S_X)$. For Kraus operators T_k of $\bar{\Xi}$, (C) gives $\sum_k T_k^\dagger \Pi' T_k \le \Pi$, hence $\Pi' T_k = \Pi' T_k \Pi$ and $\Pi' \bar{\Xi}(Z) \Pi' = \Phi(\Pi Z \Pi)$ with the completely positive map $\Phi(Y) = \sum_k \Pi' T_k Y T_k^\dagger \Pi'$. Since $\Pi S_X \Pi \le B_\sigma(w)(X)\, \Pi$ and $\Phi(\Pi) \le \Pi' \bar{\Xi}(\mathbb{I}) \Pi'$, (D) gives

$$\Pi' S_Y \Pi' = \tfrac{M'}{M} \Phi(\Pi S_X \Pi) \le B_\sigma(w)(X)\, \tfrac{M'}{M} \Pi' \bar{\Xi}(\mathbb{I}) \Pi' \le B_\sigma(w)(X)\, \Pi'.$$

□

Proposition 9 (Category)**.** *Skin data and free morphisms form a category. With the tensor product $X \otimes Y$ (Hilbert space $H_X \otimes H_Y$, responses (m, n), states $\rho^h_m \otimes \rho^h_n$, collars $\Pi^X_{\sigma,w} \otimes \Pi^Y_{\sigma,w}$) it is symmetric monoidal, and $\mathfrak{O}$ is closed under $\otimes$. The monoidal unit $\mathcal{I}$ (the space $\mathbb{C}$, one response, $\Pi_{\sigma,w} = 1$) is not boundary-separated. A morphism $\mathcal{I} \to X$ exists if and only if $\omega^R_X = \omega^L_X$ and $X \in \mathcal{F}_{\mathrm{acc}}$.*

Proof. Composition: $(\mathcal{K}_2 \otimes \Xi_2)(\mathcal{K}_1 \otimes \Xi_1) = \mathcal{K}_2 \mathcal{K}_1 \otimes \Xi_2 \Xi_1$, positivity gives $(\Xi_2 \Xi_1)^\dagger(\Pi'') \le \Xi_1^\dagger(\Pi') \le \Pi$, uniform weights are preserved, and with Φ_2 as in the proof of Lemma 8,

$$\tfrac{M''}{M} \Pi'' \bar{\Xi}_2 \bar{\Xi}_1(\mathbb{I}) \Pi'' = \tfrac{M''}{M'} \Phi_2\big(\tfrac{M'}{M} \Pi' \bar{\Xi}_1(\mathbb{I}) \Pi'\big) \le \tfrac{M''}{M'} \Phi_2(\Pi') \le \Pi''.$$

Composition is bilinear, so it preserves convex combinations, and (M1) composes. For tensor products, $A' \otimes B' - A \otimes B = (A' - A) \otimes B' + A \otimes (B' - B) \ge 0$ for $0 \le A \le A'$ and $0 \le B \le B'$ gives (C) and (D); associators and the swap satisfy both with equality, and products of orthogonal projectors are orthogonal. A free processing of $\mathcal{I}$ prepares a cq state, which (M1) forces to equal ω^R_X and ω^L_X; its average channel prepares S_X/M, so (D) reads $\Pi S_X \Pi \le \Pi$. Conversely, if $\omega^R_X = \omega^L_X$ and $X \in \mathcal{F}_{\mathrm{acc}}$, preparing ρ^R_m with label m with probability $1/M$ is free. □

Theorem 10 (No skin creation, and monotones)**.** *Let $X \to Y$.*

(i) Golden rules. *If $X \in \mathcal{F}_{\mathrm{dir}}$ then $Y \in \mathcal{F}_{\mathrm{dir}}$, and if $X \in \mathcal{F}_{\mathrm{acc}}$ then $Y \in \mathcal{F}_{\mathrm{acc}}$. Hence $\mathcal{F}$ is closed under free morphisms.*

(ii) Leakage monotones. *Let $\psi : [0,1]^{2|\Sigma|W} \to (-\infty, \infty]$ be convex and non-increasing in each coordinate, and let $\Lambda_m = (\lambda^R_m(\sigma, w), \lambda^L_m(\sigma, w))_{\sigma,w}$. Then $\Psi_\psi(X) := \mathbb{E}_\pi \psi(\Lambda_m)$ satisfies $\Psi_\psi(Y) \le \Psi_\psi(X)$. This covers $\mathcal{R}_\sigma(w)$, the mean right collar weight $c^R_\sigma(w)$, the faithful monotone $\mathcal{S}(X) := \mathbb{E}_\pi \max_{\sigma,w} s_m(\sigma, w)$ (which vanishes if and only if $X \in \mathcal{F}_{\mathrm{dir}}$), and the rates of Definition 6 along levelwise free sequences.*

(iii) Accumulation monotone. *$B_\sigma(w)(Y) \le B_\sigma(w)(X)$ for all σ, w.*

(iv) Divergence monotones. *For every function $\mathbf{D}$ on pairs of states that satisfies data processing, $\mathbf{D}(\omega^R_Y \| \omega^L_Y) \le \mathbf{D}(\omega^R_X \| \omega^L_X)$. Only (M1) is used. In particular Φ_Q is monotone and additive, $\Phi_Q(X \otimes Y) = \Phi_Q(X) + \Phi_Q(Y)$, and for generator data $\Phi_Q = -\log \mathbb{E}_\pi \kappa_m^{-1}$.*

Proof. (i) For $\mathcal{F}_{\mathrm{dir}}$, Lemma 7 gives $\lambda'^R + \lambda'^L \geq \sum_m \beta_{mm'}(\lambda^R_m + \lambda^L_m) \geq 1$. For $\mathcal{F}_{\mathrm{acc}}$, use Lemma 8. (ii) Since ψ is non-increasing, Lemma 7 and Jensen's inequality give $\psi(\Lambda'_{m'}) \leq \psi(\sum_m \beta_{mm'}\Lambda_m) \leq \sum_m \beta_{mm'}\psi(\Lambda_m)$. Multiply by $\pi'_{m'}$, sum, and use $\sum_{m'} \nu_{m'm} = 1$. The score $s = \max\{0, -\frac{1}{2}\log(\lambda^R + \lambda^L)\}$ is a maximum of convex non-increasing functions, and so is $\max_{\sigma,w} s$. Faithfulness holds because all $\pi_m > 0$. (iii) is Lemma 8. (iv) Data processing under Θ together with (M1). For cq operators with a common classical part, $F(\omega^R, \omega^L) = \sum_m \pi_m F(\rho^R_m, \rho^L_m)$, and F is multiplicative under $\otimes$. □

Theorem 11 (Completeness)**.**

(a) Leakage preorder. *For X, Y with finitely many responses the following are equivalent: (i) there is a column-stochastic ν with $\pi^Y = \nu\pi^X$ and $\Lambda^Y_{m'} \geq \sum_m \beta_{mm'}\Lambda^X_m$ coordinatewise (the conclusion of Lemma 7); (ii) $\Psi_\psi(Y) \leq \Psi_\psi(X)$ for all ψ as in Theorem 10(ii); (iii) the same holds for all $\psi(\Lambda) = \max_{k \leq M_Y}(\alpha_k - c_k \cdot \Lambda)$ with $c_k \geq 0$.*

(b) Full conversion. *For a type τ and Hermitian operators $U = (U_R, U_L)$ on $\mathbb{C}^{M'} \otimes H'$, let*

$$V_{\tau,U}(X) = \max\big\{\mathrm{Tr}[U_R\,\Theta(\omega^R_X)] + \mathrm{Tr}[U_L\,\Theta(\omega^L_X)] : \ \Theta \text{ a free processing of } X \text{ into } \tau\big\},$$

with $\max\emptyset = -\infty$. Each $V_{\tau,U}$ is a monotone. It is the value of a semidefinite program with one block per deterministic relabeling $f : [M] \to [M']$ (a linear program in the commuting sector). For Y of type τ, $X \to Y$ holds if and only if $V_{\tau,U}(X) \geq V_{\tau,U}(Y)$ for every U.

Proof. (a) (i)⇒(ii) is the proof of Theorem 10(ii), and (ii)⇒(iii) is trivial. For (iii)⇒(i), (i) is a linear program in the coupling $J_{mm'} = \nu_{m'm}\pi_m \geq 0$ with marginals π, π' and $\sum_m J_{mm'}\Lambda_m \leq \pi'_{m'}\Lambda'_{m'}$. If it is infeasible, Farkas' lemma [29] gives $a_m, b_{m'} \in \mathbb{R}$ and $c_{m'} \geq 0$ with $a_m + b_{m'} + c_{m'} \cdot \Lambda_m \geq 0$ and $\sum_m \pi_m a_m + \sum_{m'} \pi'_{m'}(b_{m'} + c_{m'} \cdot \Lambda'_{m'}) < 0$. For $\psi(\Lambda) = \max_{m'}(-b_{m'} - c_{m'} \cdot \Lambda)$ this gives $\Psi_\psi(X) \leq \sum_m \pi_m a_m < \Psi_\psi(Y)$, contradicting (iii).

(b) Every column-stochastic ν is a convex combination of deterministic relabelings f. Hence the free processings of X into τ are exactly the maps $\Theta = \sum_f \mathcal{K}_f \otimes \Phi_f$, where the Φ_f are CP maps with $\Phi^\dagger_f(\mathbb{I}) = t_f\mathbb{I}$, $\Phi^\dagger_f(\Pi'_{\sigma,w}) \leq t_f\Pi_{\sigma,w}$, $t_f \geq 0$, $\sum_f t_f = 1$, $\sum_f t_f |f^{-1}(m')| = M/M'$ for every m', and $\frac{M'}{M}\Pi'_{\sigma,w}\big(\sum_f \Phi_f(\mathbb{I})\big)\Pi'_{\sigma,w} \leq \Pi'_{\sigma,w}$; given such Φ_f, $\Theta = \sum_{t_f>0} t_f\,\mathcal{K}_f \otimes (\Phi_f/t_f)$. These constraints are linear and semidefinite in the Choi operators of the Φ_f and define a compact convex set. Its image $\mathrm{Reach}_\tau(X) = \{(\Theta(\omega^R_X), \Theta(\omega^L_X))\}$ is therefore compact and convex, and $V_{\tau,U}$ is its support function, computed by the stated SDP. Composing a morphism $X \to X'$ with a free processing of X' into τ gives a free processing of X into τ (proof of Proposition 9), so $\mathrm{Reach}_\tau(X') \subseteq \mathrm{Reach}_\tau(X)$ and $V_{\tau,U}$ is monotone. The identity is a free processing of Y into its own type, so $V_{\tau,U}(Y) \geq \mathrm{Tr}[U_R\omega^R_Y] + \mathrm{Tr}[U_L\omega^L_Y]$. If $V_{\tau,U}(X) \geq V_{\tau,U}(Y)$ for all U, then (ω^R_Y, ω^L_Y) is not strictly separated from $\mathrm{Reach}_\tau(X)$, so it lies in that set, that is, $X \to Y$. The converse is monotonicity. □

Remark 4 (Approximate conversion). For any norm on pairs of operators with dual norm $\|\cdot\|_*$, minimax over the compact convex set of free processings into the type of Y gives

$$\min_\Theta \|\Theta(X) - Y\| = \max_{\|U\|_* \leq 1} \big[\mathrm{Tr}[U_R\omega^R_Y] + \mathrm{Tr}[U_L\omega^L_Y] - V_{\tau,U}(X)\big],$$

again an SDP. Each $V_{\tau,U}$ is the optimal payoff of a decision game played with boundary-constrained processing [23, 24], and for one response and without (C) and (D), Theorem 11(b) reduces to the comparison of statistical experiments [26, 27].

5 Free operations on generators

Write $X(G)$ for the generator datum of G with all eigenvalues as responses (Definition 2), and $X_{\Lambda'}(G')$ for the datum of G' restricted to a set Λ' of simple eigenvalues. A generator is *irreducible* if no subspace other than 0 and H is invariant under both G and $G^\dagger$.

Theorem 12 (Free operations on generators)**.** *Let G on H have simple spectrum.*

(a) Let $V : H \to H'$ be an isometry with $V^\dagger \Pi'_{\sigma,w} V \le \Pi_{\sigma,w}$ for all σ, w, let h be injective on $\operatorname{spec} G$, and let $G' = Vh(G)V^\dagger + QKQ$ with $Q = \mathbb{I} - VV^\dagger$ and any K, such that the eigenvalues in $\Lambda' = h(\operatorname{spec} G)$ are simple for G'. Then $\mathcal{K}_f \otimes V(\cdot)V^\dagger$, where f relabels λ_m as $h(\lambda_m)$, is a free morphism $X(G) \to X_{\Lambda'}(G')$.

(b) Conversely, let $\Theta = \sum_j p_j \mathcal{K}_j \otimes \Xi_j$ be a free morphism $X(G) \to X_{\Lambda'}(G')$. Then every ν_j is a deterministic relabeling f_j. If f_j is injective, then:

(i) the span E'_j of the right eigenvectors of G' with labels in $f_j(\operatorname{spec} G)$ reduces G';

(ii) there are an isometry V_j of H onto E'_j and a function h_j, injective on $\operatorname{spec} G$, with $G'|_{E'_j} = V_j h_j(G) V_j^\dagger$;

(iii) $\Xi_j = V_j \mathcal{D}_j(\cdot) V_j^\dagger$, where $\mathcal{D}_j$ is a channel whose Kraus operators commute with G and $G^\dagger$. In particular $\Xi_j(\rho^h_m) = V_j \rho^h_m V_j^\dagger$, and $\mathcal{D}_j = \mathrm{id}$ if G is irreducible;

(iv) $\kappa_{f_j(m)}(G') = \kappa_m(G)$ and $s_{f_j(m)}(\sigma, w) \le s_m(\sigma, w)$ for all m, σ, w.

Proof. (a) Since $QV = 0$, $G'VR_m = Vh(G)R_m = h(\lambda_m)VR_m$, and likewise $G'^\dagger VL_m = \overline{h(\lambda_m)} VL_m$. As $h(\lambda_m)$ is simple for G', these are its unit eigenvectors, which gives (M1). (C) is the hypothesis on V, and (D) holds because $VV^\dagger \le \mathbb{I}$.

(b) *Step 1.* Eigenvectors of distinct eigenvalues are linearly independent, so the target responses are pure and pairwise distinct. By (M1), $\pi'_{m'} \rho'^R_{m'} = \sum_{j,m} p_j \nu^j_{m'm} \pi_m \Xi_j(\rho^R_m)$. Pure states are extremal, so $\Xi_j(\rho^R_m) = \rho'^R_{m'}$ whenever $p_j \nu^j_{m'm} > 0$, and likewise for L. Hence each column of ν_j is supported on one label $f_j(m)$.

Step 2. Fix j with $f = f_j$ injective, and Kraus operators T_k of Ξ_j. Purity forces $T_k R_m = a_{km} R'_{f(m)}$ and $T_k L_m = b_{km} L'_{f(m)}$, with $\sum_k |a_{km}|^2 = \sum_k |b_{km}|^2 = 1$. Since the R_m span H, $\operatorname{ran} T_k \subseteq E'_R := \operatorname{span}\{R'_{f(m)}\}$, and likewise $\operatorname{ran} T_k \subseteq E'_L := \operatorname{span}\{L'_{f(m)}\}$. Every $R'_{f(m)}$ lies in some $\operatorname{ran} T_k$, so $E'_R \subseteq E'_L$, and by symmetry $E'_R = E'_L$. For the spectral projector P of G' onto the labels $f(\operatorname{spec} G)$, $\operatorname{ran} P = E'_R$ and $\operatorname{ran} P^\dagger = E'_L = (\ker P)^\perp$, so P is an orthogonal projector. It commutes with G' and, being self-adjoint, with $G'^\dagger$. So $E' := \operatorname{ran} P$ reduces G', which is (i).

Step 3. Let $G'' = \sum_m \lambda_m P'_{f(m)}$, where $P'_{m'}$ are the spectral projectors of G'. It is a function of $G'|_{E'}$, and evaluation on $\{R_m\}$ and $\{L_m\}$ gives $T_k G = G'' T_k$ and $T_k G^\dagger = G''^\dagger T_k$. Hence every $T_k^\dagger T_l$ commutes with G and $G^\dagger$.

Step 4. The $*$-algebra generated by G is $\bigoplus_i B(H_i) \otimes \mathbb{I}_{n_i}$ for an orthogonal decomposition $H = \bigoplus_i H_i \otimes \mathbb{C}^{n_i}$ [40], and simplicity of the spectrum forces $n_i = 1$. So the commutant is spanned by the projectors Z_i onto the H_i, each R_m and L_m lies in one block, and $T_k^\dagger T_l = \sum_i c^{(i)}_{kl} Z_i$. Fix i and k with $c^{(i)}_{kk} > 0$; then $V_i = T_k Z_i / (c^{(i)}_{kk})^{1/2}$ is an isometry on H_i. For R_n in block i, $T_l R_n$ and $V_i R_n$ are multiples of $R'_{f(n)}$, and the inner product with $T_k R_n$ fixes their ratio at $c^{(i)}_{kl} / (c^{(i)}_{kk})^{1/2}$, independent of n, so $T_l Z_i = \gamma_{li} V_i$. Block-diagonality of $T_k^\dagger T_l$ makes the ranges of the V_i orthogonal, so $V = \sum_i V_i Z_i$ is an isometry onto E' (by dimension), and $\Xi_j = V \mathcal{D}(\cdot) V^\dagger$, where $\mathcal{D}$ has Kraus

operators $D_k = \sum_i \gamma_{ki} Z_i$. They commute with G and $G^\dagger$, and $\mathcal{D}$ fixes each ρ_m^h. As $VR_m \propto R'_{f(m)}$, $G'|_{E'} = Vh(G)V^\dagger$ with $h(\lambda_m) = \lambda'_{f(m)}$, which gives (ii) and (iii).

Step 5. An isometry preserves $|L_m^\dagger R_m|$. By (C), $\mathrm{Tr}[\Pi' V \rho_m^h V^\dagger] = \mathrm{Tr}[\Xi_j^\dagger(\Pi')\rho_m^h] \le \mathrm{Tr}[\Pi \rho_m^h]$ for every collar. This gives (iv). □

Theorem 12 identifies the free operations on generators. One that keeps the eigenmodes distinct can only embed a function $h(G)$, such as e^{-iGt} or a resolvent, through a boundary-compatible isometry as a decoupled block of a larger generator, and dephase between the symmetry sectors of G. Condition numbers and Petermann factors are invariant, and B cannot grow. The other free operations (noise, erasure, dilution, merging of modes) act on recorded responses, not on generators.

Remark 5 (Operations that are not free).

(a) *Similarities and one-way couplings.* $G \mapsto SGS^{-1}$ sends R_m to SR_m but L_m to $S^{-\dagger}L_m$; the imaginary gauge transformation $S = \mathrm{diag}(e^{gn})$ takes the Hermitian chain ($\kappa_m = 1$) to $\mathrm{HN}_L(g)$ ($\kappa_m > 1$). For a one-way coupling $G' = \left(\begin{smallmatrix} G & 0 \\ C & K \end{smallmatrix}\right)$ with disjoint spectra, $L'_m = (L_m, 0)$ and $R'_m \propto (R_m, (\lambda_m - K)^{-1} C R_m)$, so the multiset of condition numbers grows whenever $CR_m \neq 0$ for some m, and $\mathrm{spec}\, G$ spans an invariant but non-reducing subspace. Theorem 12(b) excludes both.

(b) *Reciprocal perturbations.* Adding $\Delta = \Delta^T$ keeps a complex-symmetric generator complex-symmetric, so it never creates direction (Theorem 2). It is nevertheless not free. It can create accumulation: the complex-symmetric hopping $i\gamma$ along one axis of a Hermitian triangular lattice produces the geometry-dependent skin effect (Section 7). It can also amplify direction: for the chain adjacency matrix A and $0 < \delta < e^{-g}$, $\mathrm{HN}_L(g) - \delta A = c\,\mathrm{HN}_L(g_\delta)$ with $c = [(e^g - \delta)(e^{-g} - \delta)]^{1/2}$ and $g_\delta = \frac{1}{2}\log\frac{e^g - \delta}{e^{-g} - \delta} > g$, so by Corollary 18 the faithful monotone $\mathcal{S}$ rises by at least $(g_\delta - g)W - C_L(g) - C_L(g_\delta)$, which is positive once $W/\log L$ is large.

(c) *Mode-dependent channels.* Tagging each response, $\rho_m \mapsto \rho_m \otimes |e_m\rangle\langle e_m|$, maps $X(G)$ to the datum of $\sum_m \lambda_m P_m \otimes |e_m\rangle\langle e_m|$, whose right eigenvectors are mutually orthogonal. No free processing does this when the R_m are pairwise non-orthogonal and the κ_m are not all equal: by Step 1 each ν_j is a relabeling f_j, which is constant, $f_j \equiv c_j$, because fidelity cannot decrease and distinct labels have orthogonal targets; then $\kappa_{c_j} \le \kappa_a$ for all a, and since every label is some c_j, all κ_m would be equal.

6 Conversions

Proposition 13 (Dilution band)**.** *Let X_t be as in Proposition 6(e) with $t \le \frac{1}{2}$. Then $\lambda_m^h(X_t) = (1-t)\lambda_m^h(X) + t$, and*

$$\left| s_m(X_t) - \min\{s_m(X), \tfrac{1}{2}\log\tfrac{1}{2t}\} \right| \le \tfrac{1}{2}\log 2 \qquad \text{for all } m, \sigma, w.$$

Proof. Let $u = \lambda^R + \lambda^L$. Then $u' = (1-t)u + 2t$ satisfies $\frac{1}{2}\max(u, 2t) \le u' \le 2\max(u, 2t)$ for $t \le \frac{1}{2}$. Hence $-\frac{1}{2}\log u'$ lies within $\frac{1}{2}\log 2$ of $\min(-\frac{1}{2}\log u, \frac{1}{2}\log\frac{1}{2t})$. The map $x \mapsto x^+$ is 1-Lipschitz, and $[\min(x, c)]^+ = \min(x^+, c)$ for $c \ge 0$. □

Theorem 14 (Rate-level classification)**.** *Let (X_L) be uniformly scaling with rate r at boundary σ and scale a (Definition 6). (i) If $X_L \to Y_L$ for every L, then $r_{\sigma'}^{\pm}(Y) \le r_{\sigma'}^{\pm}(X)$ for every σ'. (ii) For every $c \in [0, r]$ there are free morphisms $X_L \to Y_L$ such that (Y_L) is uniformly scaling with rate c at σ. Hence the set of rates reachable from (X_L) is exactly $[0, r]$, and at the rate level the resource is totally ordered by the single number r.*

Proof. (i) is Theorem 10(ii) applied at each L. (ii) For $c = 0$ use erasure (Proposition 6(d)). For $c > 0$ dilute with $t_L = \frac{1}{2}e^{-2c\,a(w_L)}$. By Proposition 13, $|s_m(Y_L) - \min\{s_m(X_L), c\,a(w_L)\}| \le \frac{1}{2}\log 2$, so $s_m(Y_L)/a(w_L) \to \min(r, c) = c$ uniformly in m. □

At fixed size the order is much richer than at the rate level.

Theorem 15 (Hatano–Nelson chains are mutually inconvertible)**.** *Let* $\mathrm{HN}_L(g)$ *be the generator datum (one site per cell, uniform weights on all* L *modes) of the open chain with* $G_{n+1,n} = e^g$ *and* $G_{n,n+1} = e^{-g}$*. Let* $L \ge 2$ *and* $1 \le W \le L/2$*. If* $g \ne g'$*, then neither* $\mathrm{HN}_L(g) \to \mathrm{HN}_L(g')$ *nor* $\mathrm{HN}_L(g') \to \mathrm{HN}_L(g)$*. In particular* $\mathrm{HN}_L(g)$ *cannot be converted into the free Hermitian chain* $\mathrm{HN}_L(0)$*.*

Proof. Averaging Lemma 7 over π' shows that $X \to Y$ requires $c^R_Y(\sigma, w) \le c^R_X(\sigma, w)$ for all σ, w, where $c^R(\sigma, w) = \mathbb{E}_\pi \mathrm{Tr}[\Pi_{\sigma,w}\rho^R]$. By the similarity $G = SH_0S^{-1}$ with $S = \mathrm{diag}(e^{gn})$ (Section 7), mode j has cell law $p^{(g)}_j(n) \propto e^{2gn}\mu_j(n)$ with $\mu_j(n) = \sin^2(k_j n)$, $k_j = \pi j/(L+1)$, and $\mu_j(1) = \mu_j(L) = \sin^2 k_j > 0$. For $A = \{n \le w\}$ with $1 \le w \le L-1$, $\frac{d}{dg}p^{(g)}_j(A) = 2\,\mathrm{Cov}_g(n, \mathbf{1}_A) < 0$, because $\mathbf{1}_A$ is non-increasing in n and non-constant on the support. Hence $c^R(-, 1)$ is strictly decreasing in g and $c^R(+, 1)$ is strictly increasing. For $g > g'$ the conversion $\mathrm{HN}(g) \to \mathrm{HN}(g')$ violates the condition at $(-, 1)$. For $g < g'$ it violates it at $(+, 1)$. The reverse direction follows by exchanging g and g'. □

The obstruction is exact but can be exponentially weak:

Proposition 16 (Lower bound for approximate conversion)**.** *For every free processing* Θ *of* X *into the type of* Y*, and every* h, σ, w*,*

$$\|\Theta(\omega^h_X) - \omega^h_Y\|_1 \;\ge\; 2\,[c^h_Y(\sigma, w) - c^h_X(\sigma, w)]^+,$$

with $c^L(\sigma, w) = \mathbb{E}_\pi \mathrm{Tr}[\Pi_{\iota\sigma,w}\rho^L]$*. For* $\mathrm{HN}_L(g) \to \mathrm{HN}_L(g')$ *with* $g > g' > 0$*, the right-hand side vanishes except in the orientation* $\sigma = -$*. There it is at most* $2c^h_{g'}(-, W) \le \frac{(L+1)^2}{2(1-e^{-2g'})}e^{-2g'(L-W)}$ *for* $h = R, L$*. For* $g' = 0$ *(the Hermitian target) it equals* $2W/L$ *up to exponentially small corrections, so the Hermitian chain is not even approximately reachable when* $W \propto L$*.*

Proof. Lemma 7, averaged over π', gives $c^h_{\Theta(X)} \le c^h_X$, and $\|A\|_1 \ge 2|\mathrm{Tr}[(\mathbb{I}\otimes\Pi)A]|$ for traceless Hermitian A. For HN, $p^{(g)}_j$ is stochastically increasing in g (proof of Theorem 15), so for $g > g'$ only $\sigma = -$ contributes. There, $c^R_{g'}(-, W) = \mathbb{E}_j p^{(g')}_j(n \le W)$. The numerator is at most $\sum_{n \le W} e^{2g'(n-1)} \le e^{2g'(W-1)}/(1 - e^{-2g'})$, and the normalization is at least $e^{2g'(L-1)}\sin^2 k_j \ge 4e^{2g'(L-1)}/(L+1)^2$. The left response is the mirror image. For $g' = 0$, $\sum_j \sin^2(k_j n) = \frac{L+1}{2}$ gives $c^R_0(-, W) = W/L$. □

7 Calibration: imaginary gauge fields and edge amplitudes

Theorem 17 (Gauge bound for tridiagonal generators)**.** *Let* G *be tridiagonal on* $\ell^2(\{1, \dots, L\})$ *with real diagonal* V_n*, and off-diagonals* $t^+_n = G_{n+1,n}$*,* $t^-_n = G_{n,n+1}$ *satisfying* $t^+_n t^-_n > 0$*. Put* $\gamma_n = \frac{1}{2}\log(t^+_n/t^-_n)$ *and* $\Gamma_n = \sum_{k<n}\gamma_k$*. Let* H_0 *be the real symmetric Jacobi matrix with diagonal* V_n *and off-diagonals* $\mathrm{sgn}(t^+_n)\sqrt{t^+_n t^-_n}$*, and let* ϕ_j *be its unit eigenvectors. Then* $G = SH_0S^{-1}$ *with* $S = \mathrm{diag}(e^{\Gamma_n})$*, the spectrum is real and simple,* $R_j \propto S\phi_j$*,* $L_j \propto S^{-1}\phi_j$*,* $\phi_j(1)\phi_j(L) \ne 0$*, and for* $1 \le w \le L/2$

$$s_j(+, w) \;\ge\; \min\{\Gamma^+_w, \Gamma^-_w\} - \tfrac{1}{2}\log\frac{2}{\min\{\phi_j(1)^2, \phi_j(L)^2\}},$$

where $\Gamma_w^+ = \min_{n\le L-w}(\Gamma_L - \Gamma_n)$ *and* $\Gamma_w^- = \min_{n>w}(\Gamma_n - \Gamma_1)$. *If* $\gamma_n \ge \gamma > 0$ *for all* n, *then* $\Gamma_w^\pm \ge \gamma w$. *The mirror statement holds for* $\sigma = -$.

Proof. Conjugation by S turns the off-diagonals of H_0 into $t_n^\pm$. Since H_0 is Hermitian, $G^\dagger = S^{-1}H_0S$, so $G^\dagger S^{-1}\phi_j = \lambda_j S^{-1}\phi_j$. A Jacobi matrix with non-zero off-diagonals has simple spectrum, and its eigenvectors cannot vanish at an end, since the three-term recursion would then force $\phi_j = 0$. Now

$$\lambda_j^R(+,w) = \frac{\sum_{n\le L-w} e^{2\Gamma_n}\phi_j(n)^2}{\sum_n e^{2\Gamma_n}\phi_j(n)^2} \le \frac{e^{2(\Gamma_L-\Gamma_w^+)}\sum_n \phi_j(n)^2}{e^{2\Gamma_L}\phi_j(L)^2} = \frac{e^{-2\Gamma_w^+}}{\phi_j(L)^2},$$

because $\Gamma_n \le \Gamma_L - \Gamma_w^+$ for $n \le L-w$. Likewise $\lambda_j^L(+,w) \le e^{-2\Gamma_w^-}/\phi_j(1)^2$. Therefore $\lambda^R + \lambda^L \le 2e^{-2\min(\Gamma_w^+,\Gamma_w^-)}/\min(\phi_j(1)^2, \phi_j(L)^2)$. □

The directional score is thus the imaginary gauge flux Γ [1] accumulated toward the boundary, minus a penalty set by the edge amplitudes of the Hermitian partner H_0, for arbitrary real disorder and quasiperiodic potentials (V32-05). If H_0 is Anderson-localized, the bound becomes a competition between bias and localization length, consistent with [1, 18]; we do not derive the critical point. Scale-free [20] and critical [19] skin modes have bounded s and would need a size-relative scale a.

Corollary 18 (Hatano–Nelson, two-sided)**.** *For* $\mathrm{HN}_L(g)$ *with* $g > 0$, $L \ge 3$ *and* $1 \le w \le L/2$, *uniformly in* j,

$$|s_j(+,w) - g\,w| \le C_L(g) := \tfrac{3}{2}\log(L+1) + g + \tfrac{1}{2}\log\frac{1}{1-e^{-2g}}, \qquad s_j(-,w) = 0.$$

If $w_L \le L/2$ *and* $w_L/\log L \to \infty$, *the sequence is uniformly scaling with rate* g *at scale* $a(w) = w$ *and shows extensive directed skin accumulation with* $f = 1$. *Moreover* $c_+^R(w) \ge 1 - \epsilon_L(g,w)$ *and* $B_+(w) \ge \frac{L}{w}\big(1 - \epsilon_L(g,w)\big)$ *with* $\epsilon_L(g,w) = \frac{1}{8}(L+1)^3 e^{-2gw}$, *so the chain also shows extensive boundary accumulation when, in addition,* $w_L/L \to 0$.

Proof. Here $\gamma_n = g$ and $\phi_j(n) = \sqrt{2/(L+1)}\sin(k_j n)$, so $\phi_j(1)^2 = \phi_j(L)^2 = \frac{2}{L+1}\sin^2 k_j \ge \frac{8}{(L+1)^3}$. Theorem 17 then gives $s_j \ge gw - \frac{3}{2}\log(L+1) + \log 2$.

For the upper bound, write $p(n) \propto e^{2g(n-1)}\sin^2(k_j n)$. The normalization is at most $e^{2g(L-1)}/(1-e^{-2g})$. With $n_0 = L - w - 1 \ge 1$,

$$p(n_0) + p(n_0+1) \ge e^{-2g(w+1)}(1-e^{-2g})\big[\sin^2(k_j n_0) + \sin^2(k_j(n_0+1))\big],$$

and the bracket equals $1 - \cos k_j \cos(k_j(2n_0+1)) \ge \frac{1}{2}\sin^2 k_j \ge \frac{2}{(L+1)^2}$. Hence $\lambda^R \ge 2e^{-2g(w+1)}(1-e^{-2g})/(L+1)^2$, which gives $s_j \le g(w+1) + \log(L+1) + \frac{1}{2}\log\frac{1}{1-e^{-2g}} - \frac{1}{2}\log 2$.

For $\sigma = -$: $p^{(g)}$ is stochastically larger than $p^{(0)}$ (proof of Theorem 15), and the left law $q^{(g)}$ is the mirror image of $p^{(g)}$ under $n \mapsto L+1-n$, hence stochastically smaller than $p^{(0)}$, which is mirror-symmetric. Since $2w \le L$, both $\{n > w\}$ and $\{n \le L-w\}$ contain at least half of the mirror-symmetric weight. So $\lambda^R(-,w) = p^{(g)}(n > w) \ge p^{(0)}(n > w) \ge \frac{1}{2}$ and $\lambda^L(-,w) = q^{(g)}(n \le L-w) \ge p^{(0)}(n \le L-w) \ge \frac{1}{2}$.

For accumulation, the proof of Theorem 17 gives $\lambda_j^R(+,w) \le e^{-2gw}/\phi_j(L)^2 \le \epsilon_L(g,w)$, and $B_+(w) \ge \mathrm{Tr}[\Pi_{+,w}S_X]/w = Lc_+^R(w)/w$. □

Corollary 19 (Reciprocal accumulation)**.** *Let* $G = \mathrm{HN}_L(g) \oplus \mathrm{HN}_L(g)^T$ *with* $g > 0$, *on cells* $n = 1,\dots,L$ *with two internal levels, and* $U = \mathbb{I} \otimes i\sigma_y$. *Then* $G^T = UGU^\dagger$ *with* $U^T = -U$, *every eigenvalue of* G *is doubly degenerate and semisimple,* $X(G) \in \mathcal{F}_{\mathrm{dir}}$, *and for* $1 \le w \le L/2$ *and both boundaries* $B_\pm(w) \ge \frac{L}{w}\big(1 - \epsilon_L(g,w)\big)$.

Proof. The symmetry relations are immediate, and $\mathrm{HN}_L(g)$ and its transpose share the simple spectrum $\{\lambda_j\}$. Since $\mathrm{HN}_L(g)$ is real, the right eigenvectors of its transpose are its left eigenvectors L_j, so the right eigenspace of λ_j is spanned by $(R_j, 0)$ and $(0, L_j)$, and $S_X = \sum_j \big(|R_j\rangle\langle R_j| \oplus |L_j\rangle\langle L_j|\big)$. Theorem 2(a) gives $X(G) \in \mathcal{F}_{\mathrm{dir}}$. The collars contain both levels, so $B_+(w) \geq \big\|\Pi_{+,w} \sum_j |R_j\rangle\langle R_j| \Pi_{+,w}\big\|$, which Corollary 18 bounds. The mirror $n \mapsto L+1-n$ maps each R_j to a multiple of L_j and gives the bound at the left boundary. □

With a reciprocal antisymmetric coupling $\delta = 0.3$ between the chains, the degeneracy, $s = 0$ and the growth of B persist ($B_\pm(2) = 5.0, 8.5, 12.0$ for $L = 8, 14, 20$, V32-12): the $\mathbb{Z}_2$ skin effect [6].

Hypothesis 1 (Exponential boundary localization)**.** A family of generators G_L satisfies ($\mathrm{H}_{\mathrm{acc}}$) at boundary σ if there are $\gamma > 0$, $c \geq 0$, C and $f > 0$ such that, for a fraction at least f of the responses, $\lambda_m^R(\sigma, w) \leq CL^c e^{-2\gamma w}$ for all $w \leq W_L$. It satisfies ($\mathrm{H}_{\mathrm{dir}}$) if moreover $\lambda_m^R(\sigma, w) + \lambda_m^L(\sigma, w) \leq CL^c e^{-2\gamma w}$ on the same responses.

Under ($\mathrm{H}_{\mathrm{acc}}$), $c_\sigma^R(w) \geq f(1 - CL^c e^{-2\gamma w})$, so the family shows extensive boundary accumulation when $\gamma w_L - c \log L \to \infty$ and $\operatorname{rank} \Pi_{\sigma, w_L}/M_L \to 0$; under ($\mathrm{H}_{\mathrm{dir}}$), also $s_m \geq \gamma w - \frac{1}{2}\log(CL^c)$ on the good responses, so it shows directed skin accumulation with $a(w) = w$. Theorem 17 gives both for tridiagonal chains with bias $\gamma_n \geq \gamma > 0$ when a fraction f of modes has $\min\{\phi_j(1)^2, \phi_j(L)^2\} \geq L^{-c}$, and Corollary 19 gives ($\mathrm{H}_{\mathrm{acc}}$) without ($\mathrm{H}_{\mathrm{dir}}$). Non-Bloch band theory [2, 5, 4] predicts ($\mathrm{H}_{\mathrm{dir}}$) for banded (block-)Toeplitz generators with a non-zero point-gap winding [6, 7], up to fine-tuned exceptions with size-dependent degeneracies [32]; a non-zero spectral area predicts ($\mathrm{H}_{\mathrm{acc}}$) in two and more dimensions [8], and in reciprocal lattices ($\mathrm{H}_{\mathrm{dir}}$) fails by Theorem 2. Rigorous eigenvector decay is known only for tridiagonal chains ([33], Theorem 17) and for semi-infinite banded and k-Toeplitz operators [35, 36]; we import the rest without proof.

For the reciprocal lattice $H(\mathbf{k}) = 2\cos k_x + 2i\gamma \cos k_y$ ($\gamma = 0.6$) on the triangle $x, y \geq 0$, $x + y \leq N - 1$, with weak real disorder that lifts chiral zero modes, $|R_m|^2 = |L_m|^2$ cell by cell, so $s \equiv 0$, while B on the diagonal edge grows as 14.3, 21.9, 27.9 for $N = 12, 18, 24$ and equals 1 at $\gamma = 0$ (V32-12): geometry-dependent accumulation without direction.

8 Numerical checks

The script `nhse_qrt_v32_checks.py` (numpy/scipy, about 25 s) tests the main statements independently; tolerances are stated in the script and its output is `nhse_qrt_v32_checks_output.txt`. Eigensolvers are used only at small sizes or condition numbers below 10^6; at larger L we use the exact similarity transformation, since eigensolvers fail at pseudospectral sensitivity $\sim e^{gL}$ [12]. V32-03 and V32-07 use general weights, which contain the uniform case (Table 1).

Not tested numerically: Theorem 11(b) beyond the commuting sector (it needs an SDP solver), the category axioms, and Hypothesis 1 beyond tridiagonal chains; Theorem 12(b) is tested through Kraus- and intertwiner-space dimensions.

9 Discussion and scope

Sections 2–7 meet (U1)–(U4) for generators with semisimple spectrum, with lower bounds proved for tridiagonal chains. The following is not established here. (1) Lower bounds beyond tridiagonal chains (Hypothesis 1), in particular for finite banded matrices with complex spectra and for the geometry-dependent skin effect. (2) Exceptional points, where $\kappa_m = \infty$. (3) Maximality beyond the statements after Lemma 7, including whether reflections exchanging opposite boundaries should

Check	Statement	Result
V32-01	Lemma 1, Theorem 2	$s \le 3 \times 10^{-15}$; cell laws of R, L agree to 10^{-14}; $U^T = -U$: all eigenvalues degenerate
V32-02	Theorem 4	all bounds hold; residue identity to $3 \times 10^{-12}\kappa$; 2055 cases with $s > 1$
V32-03, 04	Theorem 10(i),(ii),(iv)	6000 leakage tests and Φ_Q tests, 0 violations; direction-free kept 600/600; Φ_Q additive to 10^{-14}
V32-05, 06	Theorem 17, Corollary 18, Remark 3	gauge slack ≥ 0.62 (2.9×10^4 cases); HN band to $L = 400$; bulk bond $\kappa \le 2.07$, $s < 1$; Hermitian $s \le 10^{-14}$
V32-07	Theorem 11(a)	LP vs. Farkas certificate 300/300; separation 197/197
V32-08	Theorem 11(b), commuting sector	reachable 37, unreachable 23, separated 23/23; on reachable pairs no monotone violation, B never grows; (D) blocks 3/20
V32-09	Props. 13, 16, Thm. 15	dilution band holds; inconvertible 120/120; LP error equals the lower bound ($4.4 \times 10^{-3} \to 1.2 \times 10^{-5}$)
V32-10	Theorem 12	embeddings exact to 10^{-14}; commutant dim. 1, 8, 2 (HN, Hermitian, two blocks) = Kraus-space dim.; non-reducing windows: none
V32-11	Remark 5	$\kappa'_m/\kappa_m = (1 + \|x_m\|^2)^{1/2}$; gauge field $\kappa_m \ge 1.91$; $\mathcal{S}$ of HN(0.3) $- 0.4A$: $2.6 \to 4.8$, $14.6 \to 25.2$; tagging excluded
V32-12	Definition 4, Corollaries 18, 19	Hermitian $B = 1$; HN $B_+(2) = 7.9, 15.5, 30.6$, bounds 140/140; $\mathbb{Z}_2$ 5.0, 8.5, 12.0 and GDSE 14.3, 21.9, 27.9 with $s = 0$; no skin $B \le 2.4$
V32-13	Lemma 8, (D)	1096 comparisons, no growth of B; coarse-graining and splitting violate (D) and give $B = 2$
V32-14	Theorems 2, 5	$\mathbb{Z}_2$: collar weights of Q^R, Q^L equal to 5×10^{-15}, multiplicities 2; Theorem 5(a),(b) hold

Table 1: Numerical checks of the main statements (`nhse_qrt_v32_checks.py`).

be free. (4) A task-based meaning of B comparable to Remark 1, and a rate-level theory of accumulation comparable to Theorem 14. (5) A generator-level form of mode merging (Theorem 12(b) assumes injective relabelings), and an SDP of polynomial size for Theorem 11(b). (6) Copy rates: Φ_Q is additive, so $X^{\otimes n} \to Y^{\otimes k}$ with $\Phi_Q(Y) > 0$ requires $k/n \le \Phi_Q(X)/\Phi_Q(Y)$; asymptotic rates are not computed. (7) Approximate conversion within the HN family for full quantum data (in the commuting sector the minimal error equals the bound of Proposition 16, V32-09). (8) Stability and interactions: under unstructured noise of size $N^{-\gamma}$, $\gamma > 1$, banded Toeplitz eigenvectors localize only on a scale $N/\log N$ [37], so directional rates are fragile, unlike under structured disorder (Theorem 17, [34]); interacting generators are not calibrated.

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